

Zero-Shot Coverage of Unseen Substance Classes for SUD Treatment-Dropout Prediction on 6.7 Million Treatment Episodes

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Abstract—Predicting substance-use-disorder (SUD) treatment dropout is well studied, yet models are typically trained one substance at a time, leaving small or emerging classes uncovered. We ask whether a single model trained only on the six highest-volume substances can predict usefully for classes it has never seen: zero-shot in the sense that no training episode belongs to the target class. Using 6.72 million episodes from the U.S. Treatment Episode Data Set (TEDS-D, 2015–2022), we train a pooled gradient-boosted tree on 15 intake-time features and evaluate it on twelve held-out small classes. It achieves mean AUROC 0.777 (range 0.745–0.806) on unseen classes, nearly matching performance on trained substances, while pooling yields no accuracy gain over substance-specific training. On those unseen classes it stays calibrated (mean decile gap 0.015) and clinically useful (positive net benefit to threshold 0.80), loses more than the trained substances across the COVID-19 discontinuity (0.051 versus 0.017), and audits clean by subgroup except at age 65+, where predicted risk exceeds observed by 0.140. TabPFN-3 is more sample-efficient at matched budgets, yet at each model’s maximum training size both converge to a shared ~ 0.78 AUROC ceiling, locating the limit in the intake features rather than the learner. One deployable model therefore covers the substance long tail without per-class retraining.

Index Terms—substance use disorder, treatment retention, machine learning, gradient boosting, zero-shot generalization, tabular foundation models, TabPFN, health informatics

I. INTRODUCTION

Retaining patients in substance-use-disorder (SUD) treatment is one of the strongest modifiable predictors of recovery, and dropout is common: in the national data used here, 39.8% of resolved episodes end in the patient leaving against professional advice. If the episodes at highest risk of dropout could be flagged at intake, programs could target limited follow-up and retention resources more efficiently, an application in which machine learning has shown broad promise across psychiatry [1].

Machine-learning models that predict SUD treatment completion or dropout from administrative intake data are well established, including on TEDS-D [2]–[5], but essentially all of this work trains and evaluates *one substance at a time* or on a single mixed cohort, and never asks whether a model generalizes to a substance class absent from training. National treatment systems see a heavy-tailed substance distribution: a handful of high-volume substances dominate, while dozens of smaller classes (and periodically new ones such as novel synthetic opioids) individually have far too few episodes to

train a reliable substance-specific model, so a per-substance paradigm leaves them uncovered.

We therefore invert the usual question, asking not “how accurately can we predict dropout for substance X ?” but **can one model, trained only on the data-rich substances, produce useful dropout predictions for the substance classes it has never seen?**

Our contributions are:

- 1) **Single-model coverage of unseen substance classes (primary).** A pooled model trained on six substances predicts dropout for twelve held-out small classes at mean AUROC 0.777 with no substance-specific training, and we test calibration, clinical utility, temporal robustness and subgroup fairness *on those unseen classes*.
- 2) **Evidence that coverage, not accuracy, is the reason to pool.** Pooled training does not improve on substance-specific training, so the pooled model earns its place by covering classes a per-substance approach cannot reach rather than by scoring higher on the ones it can.
- 3) **A two-protocol tabular-foundation-model benchmark.** We benchmark TabPFN-3 [6] against gradient boosting at matched training size and at each model’s maximum training size, separating sample efficiency from attainable accuracy and exposing a task-imposed ceiling.

II. RELATED WORK

Dropout prediction on SUD administrative data. On TEDS, Baird et al. [2] used a virtual-twins random forest to study disparities in completion, Rabinowitz et al. [4] predicted attrition in one cohort ($N \approx 30k$) at AUROC 0.63–0.67, and Nasir et al. [3] reported AUROC ≈ 0.89 for completion by joining discharge records with county-crime data. A systematic review [5] flags the shared gap: validation within a single substance or cohort, with out-of-distribution generalization rarely tested.

Transfer across substances and tabular foundation models. The closest precedent is Bailey and DeFulio [7], who transferred a convolutional network’s weights from a heroin dataset to a smaller opioid dataset, one source→target pair between two opioids, framed as a transfer method for an accuracy gain. Our design differs in scope and conclusion: twelve held-out classes spanning sedatives, stimulants and

hallucinogens, evaluated with no weight adaptation, and the *opposite* headline, that pooling does not raise accuracy, so the value is coverage. Separately, TabPFN [6] is a transformer pretrained to perform Bayesian inference over tabular datasets in one forward pass, strong in the low-data regime; it has been applied to clinical prediction but, to our knowledge, not to SUD treatment outcomes and not benchmarked against gradient boosting on a national administrative dataset.

III. DATA AND METHODS

A. Dataset

We use the SAMHSA Treatment Episode Data Set, Discharges (TEDS-D), a public, de-identified, national census of discharges from publicly funded SUD treatment facilities, pooling seven annual files (2015–2020, 2022). The 2021 file was excluded because it ships text-labeled categories rather than the numeric codes used in every other year, and a hand-built crosswalk would carry miscoding risk; the retained years span pre-COVID, COVID-onset and post-COVID periods. After restricting to episodes with a resolvable outcome and a mappable primary substance, the analytic sample is **6,717,226 episodes** with an overall dropout rate of 0.398.

B. Task definition

The prediction target is binary: *dropout* (discharge reason “left against professional advice”) versus *treatment completed*. Episodes with other discharge reasons (transfer, incarceration, death, facility termination) are excluded, leaving a clean contrast.

Predictions use the **15 intake-time features** in Table I, all recorded at admission. The primary substance (SUB1) is only the grouping variable defining the coverage split and is **never a feature**. To prevent leakage we exclude the discharge reason, all discharge (“_D”) fields, length of stay and the 18 substance flags that re-encode SUB1. In a single-year sensitivity check (2020 held out), expanding the panel to 27 pre-admission features improved pooled AUROC by only +0.005 (0.820 to 0.825), within confidence-interval width, so the 15-feature set was retained.

C. Cohort and substance split

Table II summarizes the cohort across all eighteen primary-substance classes and the demographic strata. The six “big-6” substances have enough episodes to train a substance-specific model; the remaining twelve “small” classes are the coverage targets and together are only 3.2% of episodes. The smallest big-6 substance (cocaine, 376k) is 4.9× larger than the largest small class (benzodiazepines, 76k). Base rates vary far more across substances (0.170 to 0.531) than the cohort-wide 0.398 suggests, so one model must span markedly different populations.

D. Models and evaluation

The primary model is a histogram-based gradient-boosted-tree classifier (GBT) [8], [9], `HistGradientBoostingClassifier` from `scikit-learn`

TABLE I

THE 15 INTAKE-TIME PREDICTORS. ALL ARE RECORDED AT ADMISSION, SO A PREDICTION IS AVAILABLE BEFORE TREATMENT BEGINS. SUB1 (PRIMARY SUBSTANCE) SERVES ONLY AS THE GROUPING VARIABLE DEFINING TRAINING STRATEGIES AND IS NEVER A MODEL INPUT.

TEDS field	Domain	Definition
SUB2, SUB3	Substance	Secondary and tertiary substances at admission
ROUTE1	Substance	Route of administration (oral, smoking, inhalation, injection)
FREQ1	Substance	Frequency of use of the primary substance
FRSTUSE1	Substance	Age at first use of the primary substance
SERVICES	Treatment	Service setting (detox, residential, ambulatory)
DSMCRT	Clinical	DSM diagnostic category
NOPRIOR	Treatment	Number of prior treatment episodes
PSOURCE	Referral	Referral source (self, provider, criminal justice)
AGE, GENDER	Demographic	Age band (12–14 to 65+) and reported sex
RACE	Demographic	Reported race/ethnicity
EDUC, EMPLOY	Demographic	Education completed and employment status
STFIPS	Geography	Reporting state (FIPS code)

[10]; the linear baseline is `LogisticRegression` after a `ColumnTransformer` one-hot encoding the 15 features. Hyperparameters are given in the released code. Histogram binning [11] keeps refitting across six substances, two strategies and hundreds of bootstrap resamples affordable on CPU, and handles missing values natively. `XGBClassifier` [9] at matched settings on the two largest substances agrees to within 0.002 AUROC, so no conclusion depends on the boosting library.

All 15 TEDS fields are integer-coded categoricals. GBT receives these codes directly, since binning is invariant to nominal-code ordering, whereas logistic regression needs the one-hot expansion. No scaling, imputation, or class rebalancing is applied. We compare two training strategies that differ only in which episodes enter training: *substance-specific*, fitted and tested within a single substance, and *pooled*, fitted once on all six high-volume substances and applied everywhere.

Training assembly. Every cross-sectional result comes from one fitted model. The 6,501,243 big-6 episodes are split 70/30 stratified jointly on primary substance and outcome, giving 4,550,870 training and 1,950,373 held-out episodes, and the training side is then reduced to 999,999 by subsampling on the same joint key. A single GBT is fitted on that cohort and scored, without refitting, both on the big-6 holdout and on all 215,983 unseen-class episodes. The temporal arms are year-restricted refits of the identical specification, training filtered to 2015–2018 for the COVID arm and to years before 2022 for the temporal-coverage arm. No unseen-class episode enters any training set, which is what makes the twelve held-out classes out-of-sample by construction.

We use *zero-shot* in the literal sense that the model sees zero training episodes of the target substance class; the target (dropout versus completion) is unchanged, so the shift at test time is in the covariate distribution rather than the label space. Because substance is never a model input, the fitted model

TABLE II

COHORT CHARACTERISTICS OF THE ANALYTIC SAMPLE ($N = 6,717,226$ DISCHARGE EPISODES, 2015–2022, 2021 EXCLUDED). % IS THE COLUMN PERCENTAGE OF THE FULL COHORT AND *DROP* THE WITHIN-GROUP DROPOUT RATE; THE COHORT-WIDE RATE IS 0.398. THE SIX HIGHEST-VOLUME SUBSTANCES (ALCOHOL THROUGH COCAINE) FORM THE BIG-6 TRAINING POOL AND THE REMAINING TWELVE ARE THE HELD-OUT COVERAGE TARGETS. RACE IS MISSING FOR 2.1% OF EPISODES AND SEX FOR 0.04%; AGE BAND IS COMPLETE.

Group	<i>n</i>	%	Drop.	Group	<i>n</i>	%	Drop.	Group	<i>n</i>	%	Drop.
<i>Primary substance</i>				<i>Age band at admission</i>				<i>Race/ethnicity</i>			
Alcohol	2,491,184	37.1	0.295	12-14	35,002	0.5	0.426	Alaska Native	16,029	0.2	0.288
Heroin	1,560,370	23.2	0.474	15-17	182,273	2.7	0.432	American Indian	162,380	2.4	0.319
Cannabis	840,885	12.5	0.456	18-20	215,916	3.2	0.463	Asian/Pacific Islander	2,487	0.0	0.349
Methamphetamine	776,204	11.6	0.444	21-24	588,001	8.8	0.434	Black/African American	1,190,339	17.7	0.425
Other opioids	456,633	6.8	0.473	25-29	1,119,941	16.7	0.427	White	4,374,536	65.1	0.390
Cocaine	375,967	5.6	0.442	30-34	1,108,153	16.5	0.417	Asian	47,345	0.7	0.385
Benzodiazepines	76,380	1.1	0.334	35-39	903,291	13.4	0.407	Other single race	602,916	9.0	0.448
Other amphetamines	39,015	0.6	0.421	40-44	662,231	9.9	0.392	Two or more races	147,044	2.2	0.459
OTC	37,578	0.6	0.531	45-49	606,764	9.0	0.370	Native Hawaiian/Pacific Islander	32,251	0.5	0.394
PCP	18,098	0.3	0.401	50-54	577,785	8.6	0.347	Missing	141,899	2.1	0.221
Non rx methadone	9,867	0.1	0.485	55-64	625,069	9.3	0.323	<i>Discharge year</i>			
Other hallucinogens	8,096	0.1	0.398	65+	92,800	1.4	0.290	2015	986,614	14.7	0.383
Barbiturates	7,695	0.1	0.430	<i>Sex</i>				2016	1,015,960	15.1	0.388
Other stimulants	7,624	0.1	0.402	Male	4,492,019	66.9	0.378	2017	1,080,620	16.1	0.383
Other tranquilizers	5,187	0.1	0.170	Female	2,222,621	33.1	0.437	2018	1,034,079	15.4	0.386
Other sedatives hypnotics	3,214	0.0	0.399	<i>Substance group</i>				2019	1,042,783	15.5	0.406
Inhalants	2,607	0.0	0.427	Big-6 (trainable)	6,501,243	96.8	0.397	2020	782,004	11.6	0.426
Other sedatives	622	0.0	0.476	Small (held out)	215,983	3.2	0.404	2022	775,166	11.5	0.425

applies to a held-out class unmodified. All AUROCs carry bootstrap 95% confidence intervals, and beyond discrimination we assess calibration [12] and clinical utility via decision-curve analysis [13]. GBT and logistic experiments ran on one 48-core CPU machine; TabPFN-3 additionally used one NVIDIA RTX 2000 Ada GPU.

We also benchmark **TabPFN-3**, a pretrained tabular foundation model, under two protocols. It predicts by in-context learning rather than parameter fitting, so cost scales with (train $\times$ test) size and the usable context is bounded by compute rather than data availability: $\sim 10,000$ rows on CPU versus 100,000 rows in 38.9 s on one GPU, a 60 $\times$ speedup at matched size. Under *matched training size* both models receive the same 10,000 training and 2,000 test rows, isolating sample efficiency; under *maximum training size* each receives as much as it can use (100,000 rows for TabPFN-3, up to 10⁶ for GBT), which asks which model to deploy. Both protocols score identical test rows per comparison, so differences are paired, with 95% bootstrap intervals from 300 resamples.

IV. RESULTS

A. Single-model coverage of unseen substances (primary result)

Figure 1 shows the headline result. The pooled model, trained only on the six big-6 substances, predicts dropout for each of the twelve held-out small classes at a mean AUROC of **0.777** (unweighted; range 0.745–0.806), so every small class is predictable despite contributing nothing to training. Performance is not uniform: six classes sit above the big-6 internal reference (0.777) and six below, the lowest being benzodiazepines (0.745), barbiturates (0.748) and OTC (0.751). No class collapses toward chance. A temporal version of the test (train on years <2022, evaluate on 2022 small-class episodes) gives a similar mean AUROC of 0.785, so coverage is not an artifact of same-period evaluation.

Pooled model covers 12 unseen substance classes

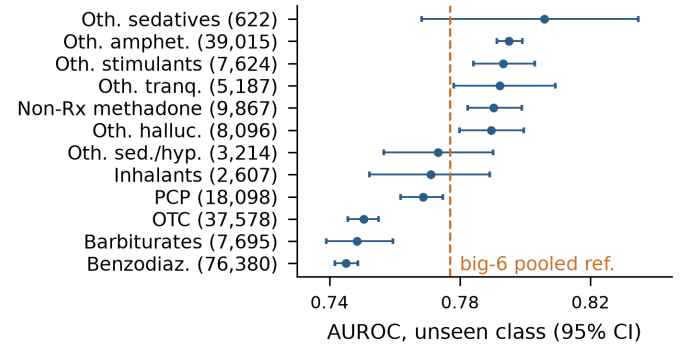


Fig. 1. **Coverage.** A pooled model trained only on the six big-6 substances predicts dropout for twelve held-out small classes it never saw (mean AUROC 0.777; dashed line = big-6 internal reference). Bars are bootstrap 95% CIs.

B. Pooling costs no accuracy

For GBT, mean AUROC across the six highest-volume substances is 0.785 with a model trained on the target substance alone and 0.776 with the pooled model, a difference of -0.008 . The penalty is small and consistent rather than concentrated in one substance: pooled training loses between 0.004 (Alcohol) and 0.012 (Other opioids) AUROC. As a supplementary analysis, rebuilding the pooled cohort with an equal number of episodes per substance rather than matching the population mixture moves per-substance AUROC by at most 0.004 (mean $+0.001$; it helps the four smaller substances and costs the two largest), so the alcohol-heavy mixture is not the source of the pooling cost. Pooling therefore does *not* beat substance-specific training, and the pooled model’s value is coverage and single-model deployment simplicity rather than an accuracy gain. Logistic regression reproduces the same ordering at a uniformly lower level (0.753 and 0.742), trailing

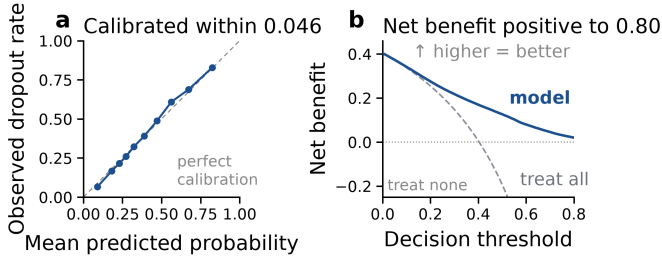


Fig. 2. **Calibration and clinical utility on the twelve unseen classes** ($n = 215,983$; pooled GBT trained only on the big-6). (a) Predicted versus observed dropout by risk decile: the ten points are deciles of predicted risk pooled across all twelve classes (about 21,600 episodes each), not individual substances, and the dashed diagonal is the identity line on which predicted equals observed. Mean absolute deviation 0.015. (b) Decision curve: net benefit stays above both defaults through threshold 0.80, while “treat all” turns negative at 0.41.

GBT by 0.031 to 0.034 AUROC, which indicates nonlinear feature interactions matter.

C. Calibration and clinical utility on unseen classes

A risk score is actionable only if its probabilities mean what they say and if acting on them beats the default policy. On the big-6 holdout, our reference, the pooled GBT reaches AUROC 0.791 and Brier 0.179.

Scoring the same model on all 215,983 unseen-class episodes gives AUROC 0.778 and Brier 0.186, and calibration transfers essentially intact (Fig. 2a): across risk deciles the mean absolute gap between predicted and observed dropout is 0.015 and never exceeds 0.046, despite the unseen cohort’s higher base rate (0.404 versus 0.397) and different substance mix. The model is therefore usable as a probability, not merely a ranking, on classes it never saw. Decision-curve analysis (Fig. 2b) shows net benefit above both defaults through threshold 0.80: 0.275 at 0.20 and 0.126 at 0.50, where “treat all” has already turned negative (crossing zero at 0.41). The margin over “treat all” widens above 0.15, the regime that matters for a program supporting part of its caseload.

D. COVID-aware temporal transportability

A model deployed on emerging substances also runs in a later era than the one it trained in, so temporal and substance shift compound. The dropout base rate rises from ~ 0.39 pre-2020 to 0.427 at COVID onset and stays elevated (0.426 in 2022), making the pandemic a natural shift test. Training on 2015–2018 only, we hold 2019 out and evaluate on 2019, 2020 and 2022 (Fig. 3).

On the trained substances the model is close to era-invariant: AUROC 0.786, 0.769, and 0.768: a one-time step of 0.017 at the pandemic boundary and no further erosion. Repeating the design end-to-end within the unseen classes (training on 125,912 unseen-class episodes from 2015–2018, testing on 35,424 / 26,861 / 27,786) gives 0.789, 0.738, and 0.708: a 0.051 step at onset and a further 0.030 by 2022, against 0.017 and 0.001. Combined substance and temporal shift therefore costs about 0.081 AUROC over seven years, whereas either alone costs under 0.020. Coverage survives the pandemic

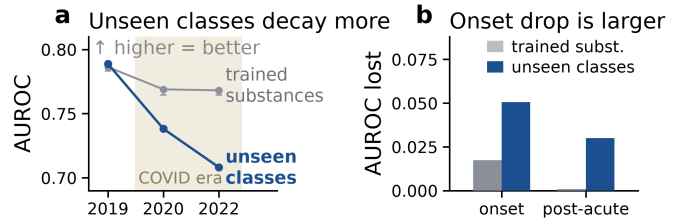


Fig. 3. **Temporal transportability across the COVID discontinuity.** Both models are trained on 2015–2018 only, 2019 held out cleanly. (a) Test-year AUROC; error bars are bootstrap 95% CIs (big-6 only, where per-year test sets exceed 700k episodes). (b) AUROC lost between consecutive eras: the unseen classes lose more at onset (0.051 versus 0.017) and keep decaying afterwards (0.030 versus 0.001).

break but decays with time in a way same-period evaluation would not reveal, arguing for periodic recalibration rather than indefinite deployment.

E. Subgroup fairness

Because algorithmic risk scores can encode and amplify demographic bias [14], we audit the model across subgroups. On the trained substances the audit is clean: calibration gaps stay within ± 0.022 everywhere and age shows a monotonic gradient (0.728 at ages 12–14 to 0.837 at 65+).

The audit that matters for our claim is on the unseen classes, across 22 race, sex and age strata (Fig. 4). AUROC spans 0.677–0.798 (standard deviation 0.032), sex is near-equal (male 0.775, female 0.780), age climbs from 0.707 at 12–14 to 0.795 at 55–64, and the lowest values fall on the smallest strata (Alaska Native, $n=260$).

Calibration on unseen classes is where the audit finds a real exception. Twenty-one of 22 strata stay within 0.045 of their observed rate, but the 65+ band predicts risk above observed by **0.140** ($n=5,061$, observed dropout 0.165 against a cohort rate of 0.404), so a threshold-based referral rule would over-flag this band. The gap is localised rather than an age effect: a single state’s other-tranquilizer episodes at 65+ ($n=3,234$, 1.5% of the cohort) report observed dropout 0.067 against 0.277 for big-6 patients of the same age in the same state, a pattern absent from the training data; excluding them the 65+ band instead under-predicts slightly, by 0.010, and no stratum exceeds 0.05. Across the eight strata with $n > 20,000$, calibration error never exceeds 0.017 and AUROC stays in 0.772–0.798, so the model is fair in discrimination and adequately calibrated for the bulk of the population.

F. Tabular foundation model benchmark

The two protocols give different answers, and the contrast is the result. At a matched 10,000/2,000-row train/test budget, TabPFN-3 beats GBT in *every* one of the 18 substance-by-strategy comparisons by a mean of +0.018 AUROC. Given each model the data it can use, that advantage disappears: TabPFN-3 at a 100,000-row context versus GBT at up to 10^6 rows wins only 8 of 18 comparisons, mean -0.0001 AUROC,

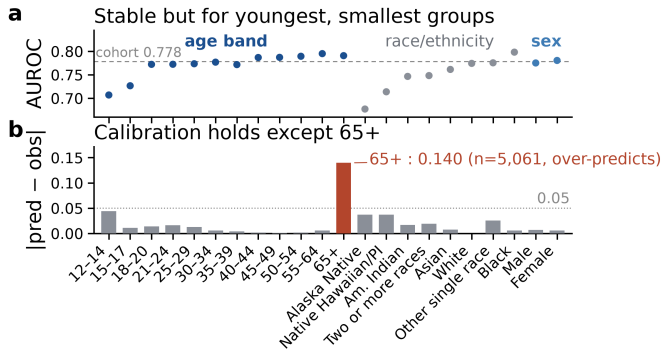


Fig. 4. **Subgroup audit on the twelve unseen classes** (22 strata; dashed line: unseen-cohort AUROC 0.778). (a) AUROC by age band, race/ethnicity, and sex; the lowest values fall on the smallest strata. (b) Absolute calibration gap per stratum: only the 65+ band (highlighted) exceeds 0.05, at 0.140, while 21 of 22 stay within 0.045.

with 15 of 18 paired bootstrap intervals including zero, per-strategy means flat, and the three significant intervals pointing both ways.

TabPFN-3’s advantage is therefore *sample efficiency*, not higher attainable accuracy. Both architectures converge near 0.78 AUROC, locating the limit in the intake-time administrative variables rather than the learner, consistent with pooling also moving accuracy very little.

V. DISCUSSION

The central finding is that a single model covers substance classes it never trained on, which is useful precisely where per-substance modeling fails: the long tail of substances too small to support their own model. The dominant predictors (treatment setting, diagnosis group, prior-treatment history) appear substance-general enough to serve all six, supported by the flat pooled-versus-specific comparison and the coverage result.

Limitations and future work. The unit is a treatment episode, not a person, and the outcome is an administrative discharge code, not a clinical assessment of recovery; 2021 is excluded. The legitimate use is population-level (ranking a caseload to target finite retention resources), while person-level deployment would require prospective, linked, IRB/DUA-governed validation. Coverage here is zero-shot generalization, the weakest form of transfer; since pooling does not beat substance-specific training we do not expect stronger adaptation to help, though a confirmatory control (importance-weighted boosting [15] or TabPFN few-shot adaptation) is future work. Two findings bound the deployment claim. The 65+ calibration gap on unseen classes (0.140) is the one audit failure, and it traces to one state-substance reporting cell rather than to age, so the indicated check is per-state label screening, not recalibration; temporal decay on unseen classes is larger than on trained substances and does not plateau, so the result licenses periodic refitting rather than indefinite use of a frozen model.

VI. CONCLUSION

On 6.72M national treatment episodes, a single pooled model predicts dropout for twelve substance classes it never saw at mean AUROC 0.777. Testing deployment assurances *on those unseen classes* rather than on the training substances, three of four hold: calibration transfers (mean decile gap 0.015), net benefit stays positive to threshold 0.80, and discrimination is stable across strata (0.677–0.798); temporal robustness is qualified, since the COVID boundary costs 0.051 AUROC on unseen classes against 0.017 on trained ones. Subject to periodic recalibration and an age-specific check, one model can cover the substance long tail, including substances that emerge after training. TabPFN-3 is the more sample-efficient learner, reaching the same accuracy band from a tenth of the episodes, but at each model’s maximum training size the two are indistinguishable, locating the limit in the intake data, not the learner.

REPRODUCIBILITY AND DATA AVAILABILITY

TEDS-D files are public from SAMHSA; all models use scikit-learn implementations and the public TabPFN-3 checkpoint. The code repository contains the analysis scripts, hyperparameters, a data dictionary and one CSV of results per experiment, including every value plotted here.

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